

Event-Resolved Gravimetric Detection of Typhoon-Driven Ocean Mass Redistribution

[PENDING: author list and data-provider collaboration]

Methods-first draft; not for submission

Abstract

We test whether named typhoons produce event-resolved ocean mass-redistribution signals in superconducting-gravimeter records after explicit separation of atmosphere, hydrology, direct ocean attraction, elastic loading, and instrument effects. The design combines gravimetry with pressure, best tracks, gridded sea level, tide gauges and GNSS, and evaluates a completely held-out typhoon together with strong-weather and quiet controls. **[PENDING: Insert data-gated multi-event results and uncertainty after station authorization.]**

1 Introduction

Previous work has modeled non-tidal ocean loading in the South China Sea and observed storm-surge loading at Helgoland. The increment tested here is not general NTOL correction, but named-typhoon attribution with a closed component budget, independent evidence, and cross-event generalization.

2 Data and preregistered event design

The raw-data gate requires either one station covering at least three typhoons or two stations covering at least two typhoons each, including gravity, colocated pressure, calibration/timing/jump/maintenance records, sea-level fields, storm tracks, precipitation, and hydrology. Evaluation additionally requires one held-out typhoon, one non-typhoon strong-weather control, three quiet windows, and no overlapping same-station windows across splits.

IBTrACS agency wind and pressure fields are retained separately. ERA5 surface pressure supplies atmospheric loading; mean-sea-level pressure and winds are event diagnostics. Significant wave height is not converted into additional water mass. CMEMS sea level is admitted only after its tide, mean dynamic topography, and inverse-barometer semantics are audited.

3 Methods

3.1 Transparent gravimeter correction

For observed gravity g_{obs} , the residual is

$$g_{\text{res}} = g_{\text{obs}} - \sum_k g_k, \quad (1)$$

where every removed component has a unique physical-effect identifier, source, SI series, and pre-applied flag. Two components cannot own the same effect. Each stage stores its input, removed series, output, and peak contribution.

Missing samples and cadence violations split records without interpolation. Candidate discontinuities are flagged at original indices but are not automatically classified or corrected.

3.2 Independent ocean forward model

Sea-level anomaly η is converted to surface density $\Delta\sigma = \rho_w\eta$. Direct attraction is integrated on spherical and WGS 84 grids with explicit masks and partial coastal-cell fractions. The modeled ocean signal is

$$g_{\text{ocean}}^{\text{model}} = g_{\text{direct}} + g_{\text{deformation}} + g_{\text{internal}}, \quad (2)$$

with every term retained separately. Elastic terms are disabled until the Green-function Earth model, reference frame, normalization, and component semantics pass a published benchmark.

3.3 Double-count audit

A CMEMS field containing inverse-barometer response cannot be combined with a second ERA5-driven inverse-barometer correction. Likewise, a pre-corrected NTOL residual cannot receive NTOL again. Unknown product physics blocks assembly.

3.4 Evaluation

Primary metrics are correlation, RMSE, peak amplitude and time error, explained variance, frequency-domain coherence, event SNR, and held-out performance. Quiet/control windows estimate false positives. Model, filter, hydrology, Green-function, station, and event sensitivities are retained, including failed events.

4 Results

A registered non-restricted readiness audit connects 17 completed processing and evaluation work units. The open IBTrACS inventory contains 48 broad regional track candidates and 11 priority station-coverage inquiries. These are not gravity events and do not select a pilot: the real-event gate remains `pending_no_event_windows`. The environmental-effect ledger also remains open because the exact historical CMEMS inverse-barometer semantics are unknown. Consequently the claim-safe manuscript branch is `pending_evidence`. **[PENDING: Insert only registered real-SG event results after the station and effect-ownership gates pass.]**

5 Discussion

Possible outcomes include successful event attribution, evidence that ocean products miss rapid coastal redistribution, or an upper constraint from non-detection. A single successful event cannot establish generality.

6 Data and code availability

Restricted SG and station logs will remain outside Git. Versioned manifests will record owner, terms, station, channels, time range, product version, retrieval time, checksum, exclusions, and redistribution limits. Processing code and compact nonrestricted metrics will be released.

7 Limitations

SG event windows, calibration and station logs remain authorization-dependent. The exact historical CMEMS pressure-response semantics are a separate product- metadata blocker. Consequently no pilot event, attribution coefficient, held-out improvement, event SNR or false-positive rate is reported. The current method audit cannot substitute for those observations.

References